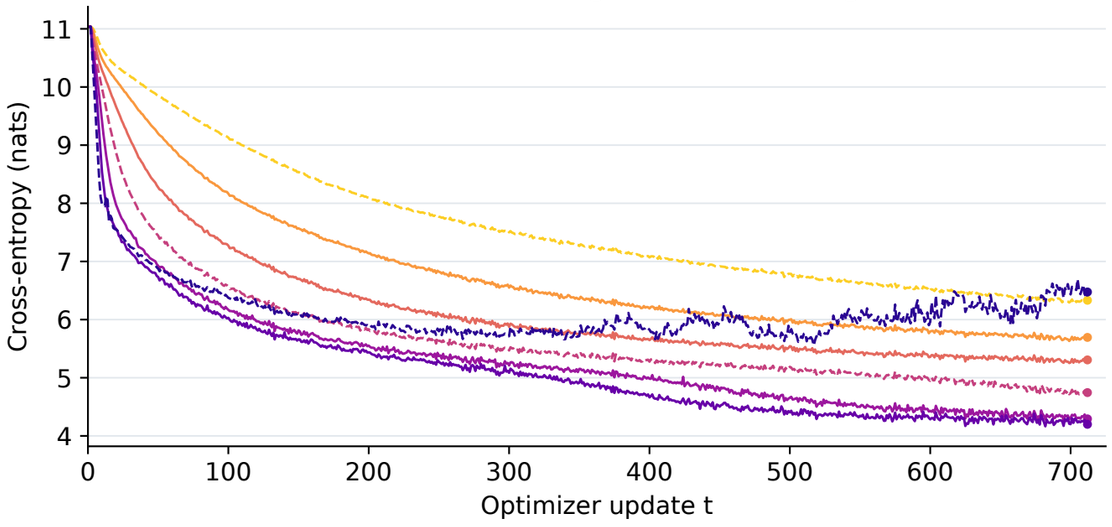
Pythia-14M · loss and raw/momentum norms by learning rate

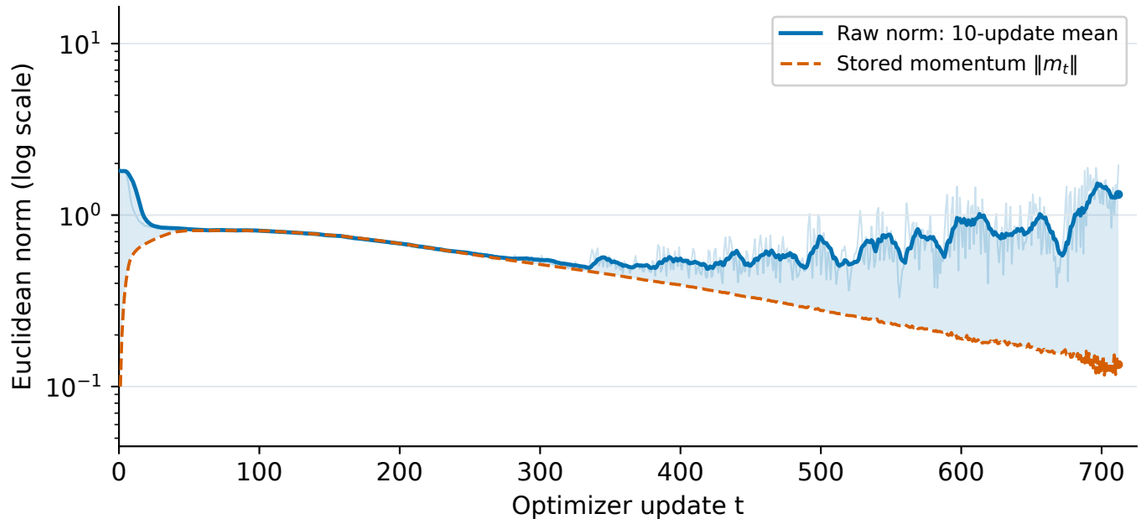
Panel (a): peak learning rate · S = Spark, R = RTX4090

0.0001 (S) · 0.000215443 (R) · 0.000464159 (R) · 0.001 (S) · 0.00215443 (R) · 0.00464159 (R) · 0.01 (S)

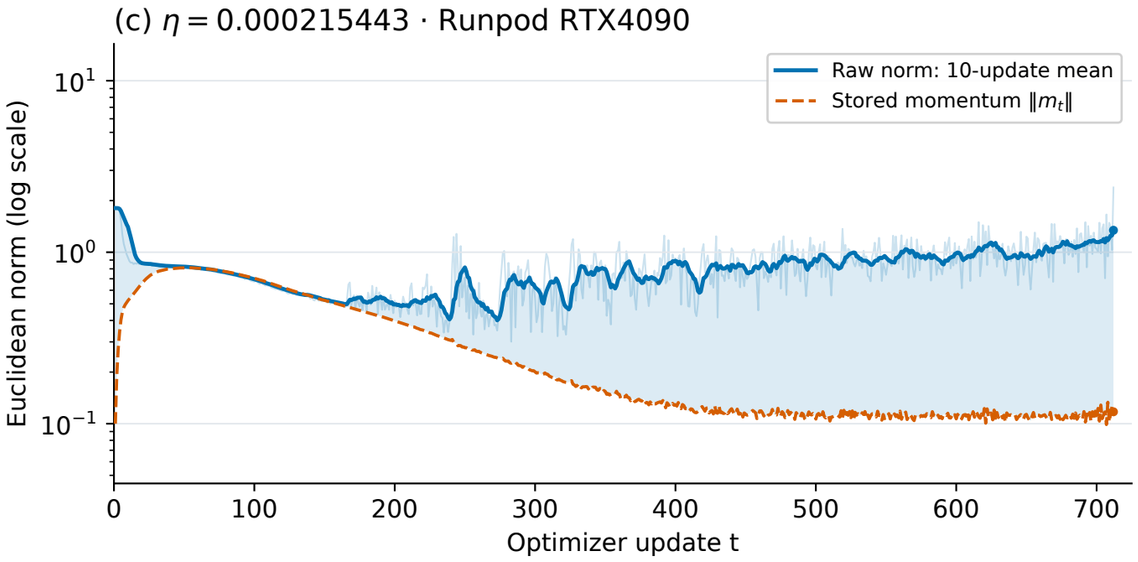
(a) Unsmoothed training loss · all learning rates



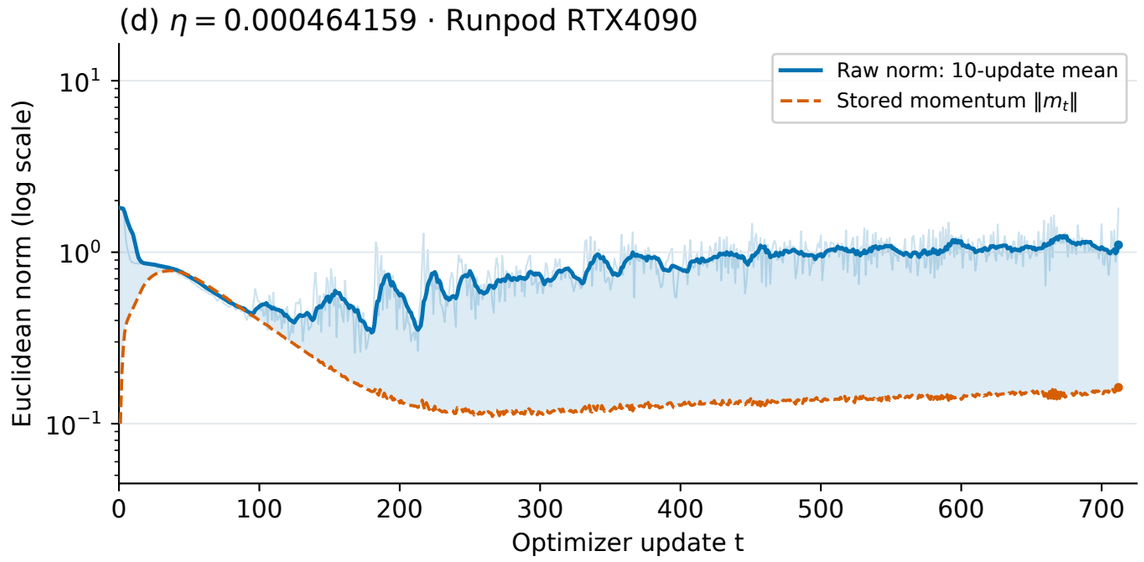
(b) $\eta = 0.0001$ · Spark



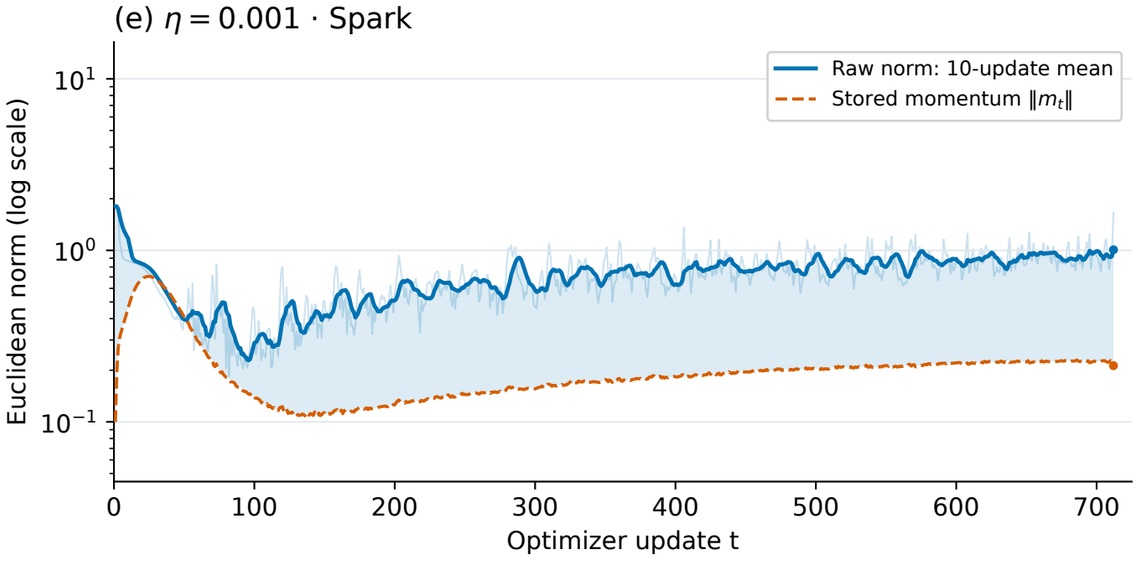
(c) $\eta = 0.000215443$ · Runpod RTX4090



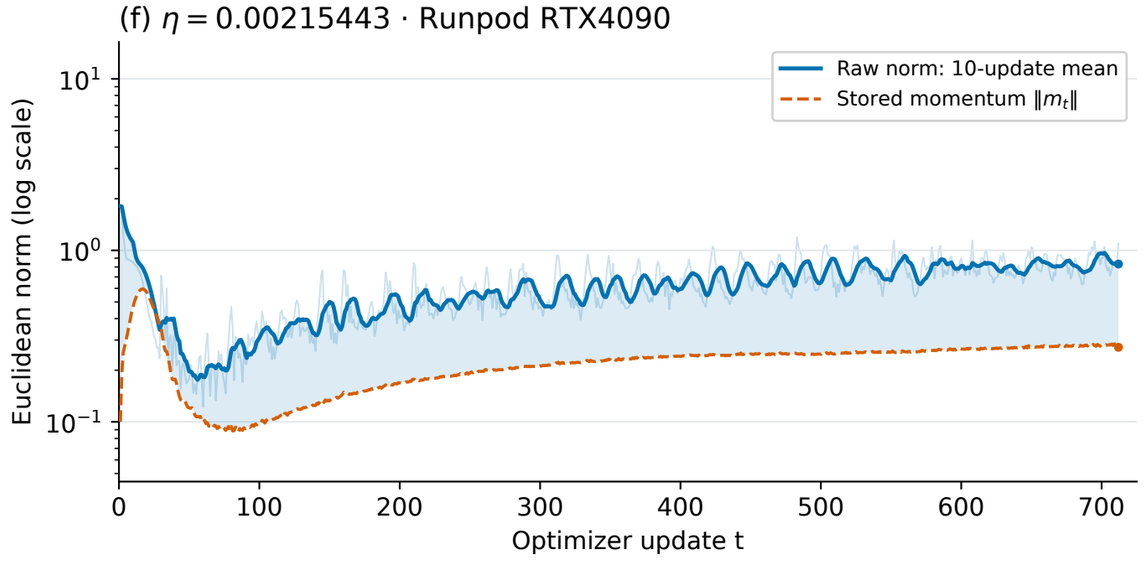
(d) $\eta = 0.000464159$ · Runpod RTX4090



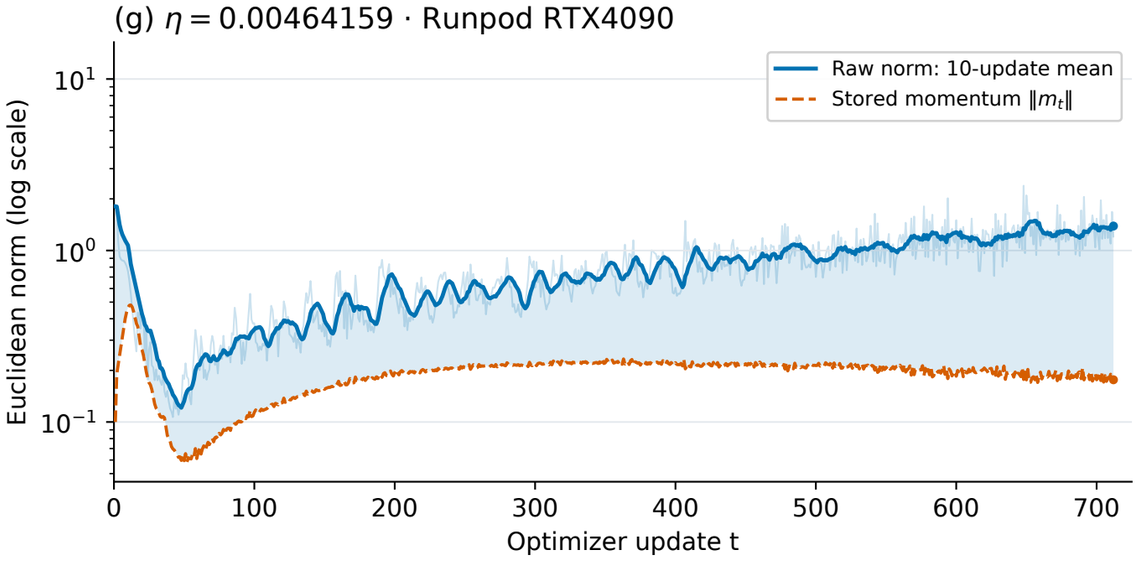
(e) $\eta = 0.001$ · Spark



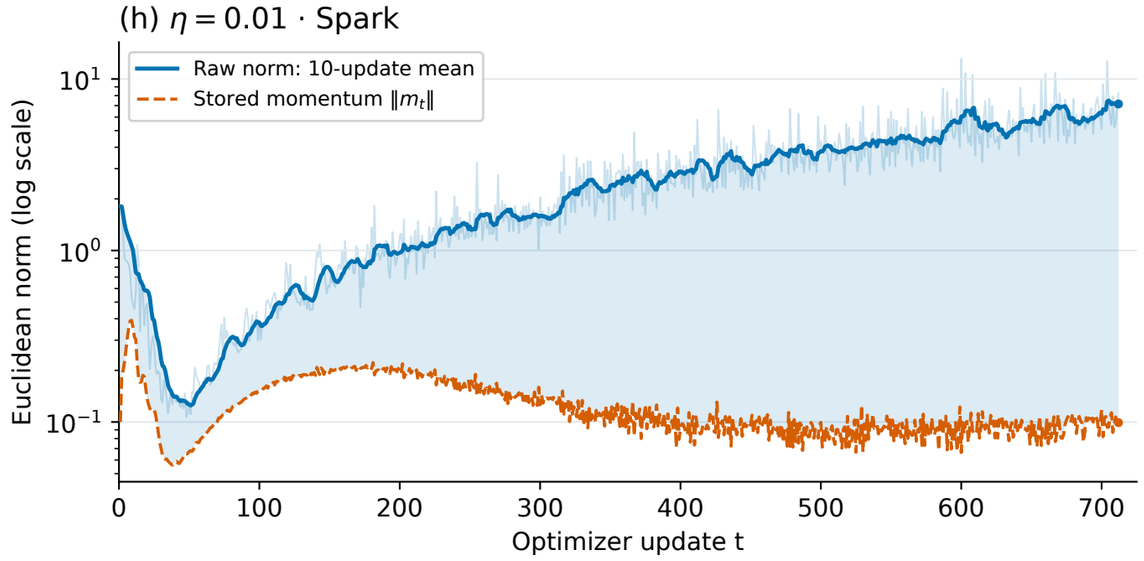
(f) $\eta = 0.00215443$ · Runpod RTX4090



(g) $\eta = 0.00464159$ · Runpod RTX4090



(h) $\eta = 0.01$ · Spark



All 712 accepted updates; loss and momentum unsmoothed. Circles mark update 712: the partial final batch (311 sequences versus 1,024).

Norm panels: faint raw values; solid blue = trailing 10-update arithmetic mean. Blue fill spans that mean and momentum; no uncertainty band.

Means use available prefix data in norm units, before log display. Norm panels share logarithmic Y limits; no normalization or positive floor.

Raw g is before clipping; stored Adam first moment m uses clipped gradients and precedes bias correction/preconditioning. Seed 1234; hardware differs across groups.